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HIGH BRIGHTNESS ZOOM PROJECTION LENS

Abstract

A high brightness zoom projection lens that meets the projection requirements of high brightness and zoom at the same time, which all lenses are made of glass lenses, satisfying $2.2 > F_t/F_w > 1.2$, F_t is the focal length at the telephoto end, F_w is the focal length at the wide-angle end, and sequentially comprising: a first lens group, having negative diopter; a second lens group, having positive diopter; a third lens group, having positive diopter; a fourth lens group, having negative diopter; a fifth lens group, having positive diopter; a sixth lens group, having positive diopter, and the Abbe number of the last lens close to the narrowing side is between 16 and 25.

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Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0001] The present invention relates to a high brightness zoom projection lens, particularly to one that meets the requirements of high brightness and zoom projection and maintain the quality of projection imaging.

2. Description of the Related Art

[0002] There is a corresponding relationship between the brightness of the projector and the brightness of the environment. When using a projector in a brighter environment, a projector with higher brightness is required. Furthermore, in addition to meeting the aforementioned requirements for high brightness projection, lens groups usually required zooming depending on the installation position of the projector to adjust the projection image to an appropriate size. Therefore, making the projection lens meet the projection requirements of high brightness and zoom at the same time, and can also take into account the projection imaging quality are the goal of the present invention.

SUMMARY OF THE INVENTION

[0003] A primary objective of the present invention is to provide a high brightness zoom projection lens which meet the projection requirements of high brightness and zoom at the same time.

[0004] To achieve the objects mentioned above, the present invention comprises: a high brightness zoom projection lens, which all lenses are made of glass lenses, satisfying $2.2 > F_t/F_w > 1.2$, F_t is the focal length at the telephoto end, F_w is the focal length at the wide-angle end, and from the magnifying side to the narrowing side of the lens sequentially comprising: [0005] a first lens group, having negative diopter, being a fixed group when zooming, and a first lens close to the magnifying side has positive diopter; a second lens group, having positive diopter, being a movable group when zooming; a third lens group, having positive diopter, being a movable group when zooming, and the amount of movement is the largest; a fourth lens group, having negative diopter, being a movable group when zooming; a fifth lens group, having positive diopter, being a movable group when zooming; a sixth lens group, having positive diopter, being a fixed group when zooming, and the Abbe number of the last lens close to the narrowing side is between 16~25.

[0006] Moreover, the first lens group **11** having a focal length value of F_1 satisfying $-1.1 > F_1/F_w > -2.5$, the second lens group **12** having a focal length value of F_2 satisfying $5.6 > F_2/F_w > 2.8$, the third lens group **13** having a focal length value of F_3 satisfying $4.2 > F_3/F_w > 2.5$, the fourth lens group **14** having a focal length value of F_4 satisfying $-1.6 > F_4/F_w > -3.1$, the fifth lens group **15** having a focal length value of F_5 satisfying $3.6 > F_5/F_w > 2.5$, the sixth lens group **16** having a focal length value of F_6 satisfying $3.6 > F_6/F_w > 2.2$. An aperture is set between the third lens group and the fourth lens group, and the F value of the aperture is between 1.7 and 2.4. The high brightness zoom projection lens satisfies $8.2 > CA/IMH > 5.8$, where CA is the effective diameter of the first lens on the magnifying side and IMH is the maximum image height on the narrowing side. The high brightness zoom projection lens satisfies $10.5 > TTL/IMH > 7.8$, where TTL is the distance from the top point of the first lens on the magnifying side to the image source on the narrowing side, and IMH is the maximum image height on the narrowing side. All lenses of the high brightness zoom projection lens are spherical lenses. The third lens group or the fourth lens group include a doublet or an achromatic doublet.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1A is a schematic diagram illustrating lenses arrangement of the wide-angle end of the first embodiment of the present invention;

[0008] FIG. 1B is a schematic diagram illustrating lenses arrangement of the telephoto end of the first embodiment of the present invention;

[0009] FIG. 1C is a graph illustrating the transverse ray fan plot of the wide-angle end of the first embodiment of the present invention;

[0010] FIG. 1D is a graph illustrating the field curvature of the wide-angle end of the first embodiment of the present invention;

[0011] FIG. 1E is a graph illustrating the distortion of the wide-angle end of the first embodiment of the present invention;

[0012] FIG. 1F is a graph illustrating the lateral color aberration of the wide-angle end of the first embodiment of the present invention;

[0013] FIG. 1G is a graph illustrating the transverse ray fan plot of the telephoto end of the first embodiment of the present invention;

[0014] FIG. 1H is a graph illustrating the field curvature of the telephoto end of the first embodiment of the present invention;

[0015] FIG. 1I is a graph illustrating the distortion of the telephoto end of the first embodiment of the present invention;

[0016] FIG. 1J is a graph illustrating the lateral color aberration of the telephoto end of the first embodiment of the present invention;

[0017] FIG. 2A is a schematic diagram illustrating lenses arrangement of the wide-angle end of the second embodiment of the present invention;

[0018] FIG. 2B is a schematic diagram illustrating lenses arrangement of the telephoto end of the second embodiment of the present invention;

[0019] FIG. 2C is a graph illustrating the transverse ray fan plot of the wide-angle end of the second embodiment of the present invention;

[0020] FIG. 2D is a graph illustrating the field curvature of the wide-angle end of the second embodiment of the present invention;

[0021] FIG. 2E is a graph illustrating the distortion of the wide-angle end of the second embodiment of the present invention;

[0022] FIG. 2F is a graph illustrating the lateral color aberration of the wide-angle end of the second embodiment of the present invention;

[0023] FIG. 2G is a graph illustrating the transverse ray fan plot of the telephoto end of the second embodiment of the present invention;

[0024] FIG. 2H is a graph illustrating the field curvature of the telephoto end of the second embodiment of the present invention;

[0025] FIG. 2I is a graph illustrating the distortion of the telephoto end of the second embodiment of the present invention;

[0026] FIG. 2J is a graph illustrating the lateral color aberration of the telephoto end of the second embodiment of the present invention;

[0027] FIG. 3A is a schematic diagram illustrating lenses arrangement of the wide-angle end of the third embodiment of the present invention;

[0028] FIG. 3B is a schematic diagram illustrating lenses arrangement of the telephoto end of the third embodiment of the present invention;

[0029] FIG. 3C is a graph illustrating the transverse ray fan plot of the wide-angle end of the third embodiment of the present invention;

[0030] FIG. 3D is a graph illustrating the field curvature of the wide-angle end of the third embodiment of the present invention;

[0031] FIG. 3E is a graph illustrating the distortion of the wide-angle end of the third embodiment of the present invention;

[0032] FIG. 3F is a graph illustrating the lateral color aberration of the wide-angle end of the third embodiment of the present invention;

[0033] FIG. 3G is a graph illustrating the transverse ray fan plot of the telephoto end of the third embodiment of the present invention;

[0034] FIG. 3H is a graph illustrating the field curvature of the telephoto end of the third embodiment of the present invention;

[0035] FIG. 3I is a graph illustrating the distortion of the telephoto end of the third embodiment of the present invention;

[0036] FIG. 3J is a graph illustrating the lateral color aberration of the telephoto end of the third embodiment of the present invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

[0037] Referring to FIGS. 1A and 1B, the first embodiment of the present invention includes a high brightness zoom projection lens **10**, which all lenses are made of glass lenses, the focal length F_t of the telephoto end is 37.43, the focal length F_w of the wide-angle end is 24.95, $F_t/F_w=1.50$, and from the magnifying side to the narrowing side of the lens sequentially comprising: [0038] a first lens group **11**, having negative diopter and a focal length value $F_1=-35.86$, $F_1/F_w=-1.44$, being a fixed group when zooming, the first lens group **11** has a first lens **1L1**, a second lens **1L2**, a third lens **1L3**, a fourth lens **1L4** and a fifth lens **1L5** in sequence, and a first lens **1L1** close to the magnifying side has positive diopter and a focal length value $FL=233.91$; [0039] a second lens group **12**, having positive diopter and a focal length value $F_2=85.08$, $F_2/F_w=3.41$, being a movable group when zooming, the second lens group **12** has a sixth lens **1L6**; [0040] a third lens group **13**, having positive diopter and a focal length value $F_3=80.14$, $F_3/F_w=3.21$, being a movable group when zooming, and the amount of movement is the largest, third lens group **13** has a seventh lens **1L7**, an eighth lens **1L8** and a ninth lens **1L9** in sequence; [0041] a fourth lens group **14**, having negative diopter and a focal length value $(F_4)=-63.41$, $F_4/F_w=-2.54$, being a movable group when zooming, and the fourth lens group **14** has a tenth lens **1L10**, an eleventh lens **1L11**, a twelfth lens **1L12** and a thirteenth lens **1L13** in sequence; [0042] a fifth lens group **15**, having positive diopter and a focal length value $(F_5)=82.70$, $F_5/F_w=3.31$, being a movable group when zooming, and the fifth lens group **15** has a fourteenth lens **1L14**; [0043] a sixth lens group **16**, having positive diopter and a focal length value $(F_6)=82.39$, $F_6/F_w=3.30$, being a fixed group when zooming, and the sixth lens group **16** has a fifteenth lens **1L15**, and the Abbe number of the fifteenth lens **1L15** (the last lens) close to the narrowing side is 17.94; and [0044] a transmissive smooth picture actuator **T** is arranged sequentially behind the fifteenth lens **1L15**, which is a glass plate device that can rotate rapidly and minutely, it synthesizes and improves the resolution through image shifting, and sequentially arranges a prism **P**, a cover glass **C** and an image source **IMA** behind the transmissive smooth picture actuator **T**.

[0045] In addition, all lenses of the high brightness zoom projection lens **10** are spherical lenses, and an aperture **S** is set between the third lens group **13** and the fourth lens group **14**, and the F value of the aperture **S** is 2.0; The effective diameter CA of the first lens **1L1** is 104.00, the distance TTL from the top point of the first lens **1L1** to the image source on the narrowing side is 242.27, and the maximum image height IMH on the narrowing side of the lens is 14.50, $CA/IMH=7.17$, $TTL/F_w=9.71$.

[0046] The lens parameters design of the high brightness zoom projection lens **10** are shown in Table 1A and Table 1B below; Wherein **1L1R1** is the magnifying side surface **R1** of the first lens **1L1**, and **1L1R2** is the narrowing side surface **R2** of the first lens **1L1**, **1L2R1** is the magnifying side surface **R1** of the second lens **1L2**, **1L2R2** is the narrowing side surface **R2** of the second lens **1L2**, . . . **1L15R1** is the magnifying side surface **R1** of the fifteenth lens **1L15**, **1L15R2** is the narrowing side surface **R2** of the fifteenth lens **1L15**, and so on; and the twelfth lens **1L12** and the thirteenth lens **1L13** can form a doublet or an achromatic doublet.

TABLE-US-00001	1A	Comment	Radius(mm)	Thickness(mm)	Nd	Vd
	1L1R1	127.44	12.28	1.62		
	60.37	1L1R2	979.57	0.20		
	1L2R1	60.29	7.90	1.85	23.79	
	1L2R2	90.67	0.10			
	1L3R1	75.23	3.50	1.62		
	60.37	1L3R2	34.08	12.20		
	1L4R1	692.82	2.60	1.83	42.73	
	1L4R2	37.02	7.97			
	1L5R1	-395.62	2.20			
	1.50	81.61	1L5R2	47.68	A	
	1L6R1	91.22	4.88	1.85	23.79	
	1L6R2	-350.48				
	B	1L7R1	-34.70	1.80		
	1.85	23.79	1L7R2	-44.36	0.10	
	1L8R1	-85.56	4.40	1.49	70.44	
	1L8R2	-39.61	0.20			
	1L9R1	85.72				

4.48 1.62 60.37 1L9R2 -253.85 C APERTURE INF 15.35 1L10R1 -2231.73 1.40 1.81 25.48
 1L10R2 52.44 0.80 1L11R1 91.69 3.84 1.73 51.49 1L11R2 -102.08 2.47 1L12R1 -30.90 1.40 1.81
 25.48 1L12R2 81.75 0.00 1L13R1 81.75 6.23 1.50 81.61 1L13R2 -37.64 D 1L14R1 208.20 8.68
 1.44 94.52 1L14R2 -43.40 E 1L15R1 97.25 5.27 1.95 17.94 1L15R2 -408.62 11.71
 TABLE-US-00002 1B Zoom Wide Tele A 29.00 22.09 B 31.60 8.19 C 1.50 18.68 D 10.10 5.87 E
 0.50 17.87

[0047] The wide-angle end of the high brightness zoom projection lens **10** uses a first wavelength λ_1 of 620 nm, a second wavelength λ_2 of 546 nm and a third wavelength λ_3 of 455 nm to simulate different transverse ray fan plot as shown in FIG. **1C**, and the image source IMA presents different image heights of 0.00 mm, 2.85 mm, 5.70 mm, 8.55 mm, 11.40 mm and 14.26 mm respectively. The symbols ey, py, ex and px respectively represent the y-axis lateral aberration, y-axis pupil height, X-axis lateral aberration, x-axis pupil height, wherein maximum scale is ± 20.000 μm , the generated aberration value is controlled within the range of -20 μm ~ 20 μm ; The field curvature diagram in FIG. **1D** has a maximum field of view of 30.187 degrees, curves T and S are respectively the tangential field curvature characteristic curve and the sagittal field curvature characteristic curve, the tangential field curvature value and sagittal field curvature value are controlled within the range of 0.00 mm~0.08 mm; The distortion diagram in FIG. **1E** has a maximum field of view of 30.187 degrees, and the distortion amount is controlled within the range of -2.4% ~ 0 ; The lateral color aberration diagram in FIG. **1F** has a maximum field of view of 14.256 mm, and using a wavelength of about 0.546 microns as a reference, the lateral color aberration value is controlled within the range of -1.8 μm ~ 3.6 μm .

[0048] The wide-angle end of the high brightness zoom projection lens **10** uses a first wavelength λ_1 of 620 nm, a second wavelength λ_2 of 546 nm and a third wavelength λ_3 of 455 nm to simulate different transverse ray fan plot as shown in FIG. **1G**, and the image source IMA presents different image heights of 0.00 mm, 2.85 mm, 5.70 mm, 8.55 mm, 11.40 mm and 14.26 mm respectively. The symbols ey, py, ex and px respectively represent the y-axis lateral aberration, y-axis pupil height, X-axis lateral aberration, x-axis pupil height, wherein maximum scale is ± 20.000 μm , the generated aberration value is controlled within the range of -70 μm ~ 70 μm ; The field curvature diagram in FIG. **1H** has a maximum field of view of 20.890 degrees, curves T and S are respectively the tangential field curvature characteristic curve and the sagittal field curvature characteristic curve, the tangential field curvature value and sagittal field curvature value are controlled within the range of -0.06 mm~ 0.03 mm; The distortion diagram in FIG. **1I** has a maximum field of view of 20.890 degrees, and the distortion amount is controlled within the range of -0.6% ~ 0 ; The lateral color aberration diagram in FIG. **1J** has a maximum field of view of 14.256 mm, and using a wavelength of about 0.546 microns as a reference, the lateral color aberration value is controlled within the range of -2.4 μm ~ 0.9 μm .

[0049] Referring to FIGS. **2A** and **2B**, the second embodiment of the present invention includes a high brightness zoom projection lens **20**, which all lenses are made of glass lenses, the focal length Ft of the telephoto end is 37.90, the focal length Fw of the wide-angle end is 25.61, Ft/Fw=1.48, and from the magnifying side to the narrowing side of the lens sequentially comprising: a first lens group **21**, having negative diopter and a focal length value F1=-37.93, F1/Fw=-1.48, being a fixed group when zooming, the first lens group **21** has a first lens **2L1**, a second lens **2L2**, a third lens **2L3**, a fourth lens **2L4** and a fifth lens **2L5** in sequence, and a first lens **2L1** close to the magnifying side has positive diopter and a focal length value FL=235.22; [0050] a second lens group **22**, having positive diopter and a focal length value F2=78.63, F2/Fw=3.07, being a movable group when zooming, the second lens group **22** has a sixth lens **2L6**; [0051] a third lens group **23**, having positive diopter and a focal length value F3=72.32, F3/Fw=2.82, being a movable group when zooming, and the amount of movement is the largest, third lens group **23** has a seventh lens **2L7**, an eighth lens **2L8** and a ninth lens **2L9** in sequence; [0052] a fourth lens group **24**, having negative diopter and a focal length value (F4)=-49.83, F4/Fw=-1.95, being a movable group when

zooming, and the fourth lens group **24** has a tenth lens **2L10**, an eleventh lens **2L11**, a twelfth lens **2L12** and a thirteenth lens **2L13** in sequence; [0053] a fifth lens group **25**, having positive diopter and a focal length value (F5)=72.74, F5/Fw=2.84, being a movable group when zooming, and the fifth lens group **25** has a fourteenth lens **2L14**; [0054] a sixth lens group **26**, having positive diopter and a focal length value (F6)=76.90, F6/Fw=3.00, being a fixed group when zooming, and the sixth lens group **26** has a fifteenth lens **2L15**, and the Abbe number of the fifteenth lens **2L15** (the last lens) close to the narrowing side is 18.90; and [0055] a transmissive smooth picture actuator T is arranged sequentially behind the fifteenth lens **2L15**, which is a glass plate device that can rotate rapidly and minutely, it synthesizes and improves the resolution through image shifting, and sequentially arranges a prism P, a cover glass C and an image source IMA behind the transmissive smooth picture actuator T.

[0056] In addition, all lenses of the high brightness zoom projection lens **20** are spherical lenses, and an aperture S is set between the third lens group **23** and the fourth lens group **24**, and the F value of the aperture S is 2.0; The effective diameter CA of the first lens **2L1** is 100.00, the distance TTL from the top point of the first lens **2L1** to the image source on the narrowing side is 220.00, and the maximum image height IMH on the narrowing side of the lens is 14.50, CA/IMH=6.390, TTL/Fw=8.59.

[0057] The lens parameters design of the high brightness zoom projection lens **20** are shown in Table 2A and Table 2B below; Wherein **2L8R1** is the magnifying side surface R1 of the first lens **2L1**, and **2L1R2** is the narrowing side surface R2 of the first lens **2L1**, **2L2R1** is the magnifying side surface R1 of the second lens **2L2**, **2L2R2** is the narrowing side surface R2 of the second lens **2L2**, . . . **2L15R1** is the magnifying side surface R1 of the fifteenth lens **2L15**, **2L15R2** is the narrowing side surface R2 of the fifteenth lens **2L15**, and so on.

[0058] However, the seventh lens **2L7** and the eighth lens **2L8** can form a doublet or an achromatic doublet, the tenth lens **2L10** and the eleventh lens **2L11** can form a doublet or an achromatic doublet, and the twelfth lens **2L12** and the thirteenth lens **2L13** can form a doublet or an achromatic doublet

TABLE-US-00003 2A Comment Radius(mm) Thickness(mm) Nd Vd 2L1R1 114.19 12.97 1.50 81.61 2L1R2 4217.39 0.20 2L2R1 51.70 8.32 1.85 23.79 2L2R2 78.50 0.29 2L3R1 51.91 1.20 1.77 49.61 2L3R2 32.55 11.17 2L4R1 310.35 1.20 1.80 46.57 2L4R2 33.61 7.99 2L5R1 -488.18 1.20 1.50 81.61 2L5R2 41.01 A 2L6R1 79.10 4.45 1.85 23.79 2L6R2 -436.45 B 2L7R1 -42.78 1.20 1.76 27.51 2L8R1 277.29 4.62 1.80 46.57 2L8R2 -46.88 0.20 2L9R1 96.78 3.59 1.80 46.57 2L9R2 -190.18 C APERTURE INF 16.53 2L10R1 -94.74 1.20 1.85 30.06 2L11R1 35.81 5.64 1.50 81.61 2L11R2 -42.57 2.08 2L12R1 -23.79 2.11 1.69 31.08 2L13R1 116.61 6.91 1.50 81.61 2L13R2 -32.73 D 2L14R1 270.44 8.08 1.50 81.61 2L14R2 -41.45 E 2L15R1 83.81 5.35 1.92 18.90 2L15R2 -487.89 9.53

TABLE-US-00004 2B Zoom Wide Tele A 28.72 20.89 B 24.38 6.20 C 0.20 15.50 D 3.76 0.20 E 0.20 13.58

[0059] The wide-angle end of the high brightness zoom projection lens **20** uses a first wavelength λ_1 of 620 nm, a second wavelength λ_2 of 546 nm and a third wavelength λ_3 of 455 nm to simulate different transverse ray fan plot as shown in FIG. 2C, and the image source IMA presents different image heights of 0.00 mm, 2.85 mm, 5.70 mm, 8.55 mm, 11.40 mm and 14.26 mm respectively. The symbols ey, py, ex and px respectively represent the y-axis lateral aberration, y-axis pupil height, X-axis lateral aberration, x-axis pupil height, wherein maximum scale is $\pm 20.000 \mu\text{m}$, the generated aberration value is controlled within the range of $-20 \mu\text{m} \sim 20 \mu\text{m}$; The field curvature diagram in FIG. 2D has a maximum field of view of 29.231 degrees, curves T and S are respectively the tangential field curvature characteristic curve and the sagittal field curvature characteristic curve, the tangential field curvature value and sagittal field curvature value are controlled within the range of $0.00 \text{ mm} \sim 0.09 \text{ mm}$; The distortion diagram in FIG. 2E has a maximum field of view of 29.231 degrees, and the distortion amount is controlled within the range

of $-1.2\% \sim 0$; The lateral color aberration diagram in FIG. 2F has a maximum field of view of 14.256 mm, and using a wavelength of about 0.546 microns as a reference, the lateral color aberration value is controlled within the range of $-1.8 \mu\text{m} \sim 3.6 \mu\text{m}$.

[0060] The wide-angle end of the high brightness zoom projection lens **20** uses a first wavelength λ_1 of 620 nm, a second wavelength λ_2 of 546 nm and a third wavelength λ_3 of 455 nm to simulate different transverse ray fan plot as shown in FIG. 2G, and the image source IMA presents different image heights of 0.00 mm, 2.85 mm, 5.70 mm, 8.55 mm, 11.40 mm and 14.26 mm respectively. The symbols ey, py, ex and px respectively represent the y-axis lateral aberration, y-axis pupil height, X-axis lateral aberration, x-axis pupil height, wherein maximum scale is $\pm 20.000 \mu\text{m}$, the generated aberration value is controlled within the range of $-12 \mu\text{m} \sim 12 \mu\text{m}$; The field curvature diagram in FIG. 2H has a maximum field of view of 20.520 degrees, curves T and S are respectively the tangential field curvature characteristic curve and the sagittal field curvature characteristic curve, the tangential field curvature value and sagittal field curvature value are controlled within the range of $-0.05 \text{ mm} \sim 0.02 \text{ mm}$; The distortion diagram in FIG. 2I has a maximum field of view of 20.520 degrees, and the distortion amount is controlled within the range of $0 \sim 0.6\%$; The lateral color aberration diagram in FIG. 2J has a maximum field of view of 14.256 mm, and using a wavelength of about 0.546 microns as a reference, the lateral color aberration value is controlled within the range of $-2.0 \mu\text{m} \sim 1.0 \mu\text{m}$.

[0061] Referring to FIGS. 3A and 3B, the third embodiment of the present invention includes a high brightness zoom projection lens **30**, which all lenses are made of glass lenses, the focal length F_t of the telephoto end is 71.60, the focal length F_w of the wide-angle end is 37.51, $F_t/F_w=1.91$, and from the magnifying side to the narrowing side of the lens sequentially comprising: [0062] a first lens group **31**, having negative diopter and a focal length value $F_1=-83.37$, $F_1/F_w=-2.22$, being a fixed group when zooming, the first lens group **31** has a first lens **3L1**, a second lens **3L2**, a third lens **3L3**, a fourth lens **3L4** and a fifth lens **3L5** in sequence, and a first lens **3L1** close to the magnifying side has positive diopter and a focal length value $FL=181.36$; [0063] a second lens group **32**, having positive diopter and a focal length value $F_2=197.34$, $F_2/F_w=5.56$, being a movable group when zooming, the second lens group **32** has a sixth lens **3L6** and a seventh lens **3L7**; [0064] a third lens group **33**, having positive diopter and a focal length value $F_3=144.67$, $F_3/F_w=3.86$, being a movable group when zooming, and the amount of movement is the largest, third lens group **33** has an eighth lens **3L8** and a ninth lens **3L9** in sequence; [0065] a fourth lens group **34**, having negative diopter and a focal length value $(F_4)=-106.26$, $F_4/F_w=-2.83$, being a movable group when zooming, and the fourth lens group **34** has a tenth lens **3L10**, an eleventh lens **3L11**, a twelfth lens **3L12** and a thirteenth lens **3L13** in sequence; [0066] a fifth lens group **35**, having positive diopter and a focal length value $(F_5)=112.58$, $F_5/F_w=3.00$, being a movable group when zooming, and the fifth lens group **35** has a fourteenth lens **3L14**; [0067] a sixth lens group **36**, having positive diopter and a focal length value $(F_6)=92.15$, $F_6/F_w=2.46$, being a fixed group when zooming, and the sixth lens group **36** has a fifteenth lens **3L15**, and the Abbe number of the fifteenth lens **3L15** (the last lens) close to the narrowing side is 17.94; and [0068] a transmissive smooth picture actuator T is arranged sequentially behind the fifteenth lens **3L15**, which is a glass plate device that can rotate rapidly and minutely, it synthesizes and improves the resolution through image shifting, and sequentially arranges a prism P, a cover glass C and an image source IMA behind the transmissive smooth picture actuator T.

[0069] In addition, all lenses of the high brightness zoom projection lens **30** are spherical lenses, and an aperture S is set between the third lens group **33** and the fourth lens group **34**, and the F value of the aperture S is 2.0; The effective diameter CA of the first lens **3L1** is 98.80, the distance TTL from the top point of the first lens **3L1** to the image source on the narrowing side is 343.00, and the maximum image height IMH on the narrowing side of the lens is 14.50, $CA/IMH=6.81$, $TTL/F_w=9.14$.

[0070] The lens parameters design of the high brightness zoom projection lens **30** are shown in

Table 3A and Table 3B below; Wherein 3L1R1 is the magnifying side surface R1 of the first lens 3L1, and 3L1R2 is the narrowing side surface R2 of the first lens 3L1, 3L2R1 is the magnifying side surface R1 of the second lens 3L2, 3L2R2 is the narrowing side surface R2 of the second lens 3L2, . . . 3L15R1 is the magnifying side surface R1 of the fifteenth lens 3L15, 3L15R2 is the narrowing side surface R2 of the fifteenth lens 3L15, and so on.

TABLE-US-00005 3A Comment Radius(mm) Thickness(mm) Nd Vd 3L1R1 123.05 11.62 1.77 49.60 3L1R2 937.13 4.60 3L2R1 94.23 10.00 1.59 61.25 3L2R2 47.96 13.31 3L3R1 -331.58 2.10 1.64 60.20 3L3R2 58.79 15.69 3L4R1 -84.65 1.85 1.55 63.37 3L4R2 201.87 20.12 3L5R1 793.61 4.80 1.95 17.94 3L5R2 -235.85 A 3L6R1 143.77 9.33 1.83 42.73 3L6R2 -143.77 2.49 3L7R1 -106.36 2.25 1.69 31.16 3L7R2 2226.97 B 3L8R1 132.99 2.25 1.67 32.17 3L8R2 90.89 1.48 3L9R1 95.68 10.11 1.59 68.34 3L9R2 -191.11 C APERTURE INF D 3L10R1 42.98 7.37 1.60 65.46 3L10R2 -110.31 0.10 3L11R1 -163.72 1.25 1.81 25.46 3L11R2 44.49 8.21 3L12R1 -36.63 1.25 1.70 30.05 3L12R2 146.79 1.01 3L13R1 1025.62 5.35 1.74 49.22 3L13R2 -53.07 E 3L14R1 122.43 7.85 1.44 94.52 3L14R2 -81.23 F 3L15R1 94.53 5.88 1.95 17.94 3L15R2 -1299.14 14.36

TABLE-US-00006 3B Zoom Wide Tele A 38.38 3.00 B 43.62 2.00 C 5.35 62.61 D 25.00 3.00 E 12.26 4.00 F 6.15 56.15

[0071] The wide-angle end of the high brightness zoom projection lens 30 uses a first wavelength λ_1 of 620 nm, a second wavelength λ_2 of 546 nm and a third wavelength λ_3 of 455 nm to simulate different transverse ray fan plot as shown in FIG. 3C, and the image source IMA presents different image heights of 0.00 mm, 2.85 mm, 5.70 mm, 8.55 mm, 11.40 mm and 14.26 mm respectively. The symbols ey, py, ex and px respectively represent the y-axis lateral aberration, y-axis pupil height, X-axis lateral aberration, x-axis pupil height, wherein maximum scale is ± 20.000 μm , the generated aberration value is controlled within the range of $-16 \mu\text{m} \sim 16 \mu\text{m}$; The field curvature diagram in FIG. 3D has a maximum field of view of 21.130 degrees, curves T and S are respectively the tangential field curvature characteristic curve and the sagittal field curvature characteristic curve, the tangential field curvature value and sagittal field curvature value are controlled within the range of $-0.03 \text{ mm} \sim 0.05 \text{ mm}$; The distortion diagram in FIG. 3E has a maximum field of view of 21.130 degrees, and the distortion amount is controlled within the range of $-2.0\% \sim 0$; The lateral color aberration diagram in FIG. 3F has a maximum field of view of 14.256 mm, and using a wavelength of about 0.546 microns as a reference, the lateral color aberration value is controlled within the range of $0 \mu\text{m} \sim 3.6 \mu\text{m}$.

[0072] The wide-angle end of the high brightness zoom projection lens 30 uses a first wavelength λ_1 of 620 nm, a second wavelength λ_2 of 546 nm and a third wavelength λ_3 of 455 nm to simulate different transverse ray fan plot as shown in FIG. 2G, and the image source IMA presents different image heights of 0.00 mm, 2.85 mm, 5.70 mm, 8.55 mm, 11.40 mm and 14.26 mm respectively. The symbols ey, py, ex and px respectively represent the y-axis lateral aberration, y-axis pupil height, X-axis lateral aberration, x-axis pupil height, wherein maximum scale is ± 20.000 μm , the generated aberration value is controlled within the range of $-30 \mu\text{m} \sim 30 \mu\text{m}$; The field curvature diagram in FIG. 3H has a maximum field of view of 11.300 degrees, curves T and S are respectively the tangential field curvature characteristic curve and the sagittal field curvature characteristic curve, the tangential field curvature value and sagittal field curvature value are controlled within the range of $-0.03 \text{ mm} \sim 0.07 \text{ mm}$; The distortion diagram in FIG. 3I has a maximum field of view of 11.300 degrees, and the distortion amount is controlled within the range of $-0.6\% \sim 0$; The lateral color aberration diagram in FIG. 3J has a maximum field of view of 14.256 mm, and using a wavelength of about 0.546 microns as a reference, the lateral color aberration value is controlled within the range of $-1.2 \mu\text{m} \sim 2.2 \mu\text{m}$.

[0073] With the feature disclosed above, all lenses of the present invention are made of glass lenses for achieving high brightness projection requirement, a second lens group 12/22/32, a third lens group 13/23/33, a fourth lens group 14/24/34, and a fifth lens group 15/25/35 are movable group for achieving zooming projection requirement, at the same time, the aberration, field curvature,

distortion and lateral color aberration of the high brightness zoom projection lenses 10/20/30 can be controlled within a smaller range when they are at the wide-angle end and photography end; therefore, the present invention has the ability to simultaneously satisfy the requirements of high brightness and zoom projection needs and maintain the quality of projection imaging.

[0074] Although particular embodiment of the invention has been described in detail for purposes of illustration, various modifications and enhancements may be made without departing from the spirit and scope of the invention. Accordingly, the invention is not to be limited except by the appended claims.

Claims

1. A high brightness zoom projection lens, which satisfying $2.2 > F_t/F_w > 1.2$, F_t is the focal length at the telephoto end, F_w is the focal length at the wide-angle end, and from the magnifying side to the narrowing side of the lens sequentially comprising: a first lens group, having negative diopter, being a fixed group when zooming; a second lens group, having positive diopter, being a movable group when zooming; a third lens group, having positive diopter, being a movable group when zooming, and the amount of movement is the largest; a fourth lens group, having negative diopter, being a movable group when zooming; a fifth lens group, having positive diopter, being a movable group when zooming; a sixth lens group, having positive diopter, being a fixed group when zooming.
 2. The high brightness zoom projection lens as claimed in claim 1, wherein an aperture is set between the third lens group and the fourth lens group, and the F value of the wide-angle end aperture is between 1.7 and 2.4.
 3. The high brightness zoom projection lens as claimed in claim 1, wherein the high brightness zoom projection lens satisfies $8.2 > CA/IMH > 5.8$, where CA is the effective diameter of the first lens on the magnifying side and IMH is the maximum image height on the narrowing side.
 4. The high brightness zoom projection lens as claimed in claim 1, wherein the high brightness zoom projection lens satisfies $10.5 > TTL/F_w > 7.8$, where TTL is the distance from the top point of the first lens on the magnifying side to the image source on the narrowing side, and F_w is the focal length of the wide-angle end.
 5. The high brightness zoom projection lens as claimed in claim 1, wherein all lenses of the high brightness zoom projection lens are glass lenses or spherical lenses.
 6. The high brightness zoom projection lens as claimed in claim 1, wherein the third lens group or the fourth lens group include a doublet or an achromatic doublet.
 7. The high brightness zoom projection lens as claimed in claim 1, wherein the first lens group having a focal length value of F_1 satisfying $-1.1 > F_1/F_w > -2.5$, the second lens group having a focal length value of F_2 satisfying $5.6 > F_2/F_w > 2.8$, the third lens group having a focal length value of F_3 satisfying $4.2 > F_3/F_w > 2.5$, the fourth lens group having a focal length value of F_4 satisfying $-1.6 > F_4/F_w > -3.1$, the fifth lens group having a focal length value of F_5 satisfying $3.6 > F_5/F_w > 2.5$, the sixth lens group having a focal length value of F_6 satisfying $3.6 > F_6/F_w > 2.2$.
 8. The high brightness zoom projection lens as claimed in claim 1, wherein the first lens close to the magnifying side has positive diopter.
 9. The high brightness zoom projection lens as claimed in claim 1, wherein the Abbe number of the last lens close to the narrowing side is between 16 and 25.
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